

HOSPITAL BED STATE MANAGEMENT SYSTEM

System Architecture:

Portals:

- Admin Portal
- Staff Portal

Components:

1) *Staff Portal*

- Login Page (Hospital_name, Staff_id, Password)
- Dashboard containing (Admit patient, My_Patients List, Vitals-Updating)

2) *Admin Portal*

- Login Page (hospital_name, admin_email, password)
- Dashboard containing (Real-time occupancies of beds/wards/icu's/waiting_queue, Queue_list, Discharge-panel, Audit-log, Analytics)

To be designed -

- 1 - Login form
- 2 - Real-time occupancy widget (show free/occupied beds)
- 3 - Waiting queue widget (show next 5 patients, why they're waiting)
- 4 - Discharge management (click to discharge, log reason)
- 5 - Audit log viewer (search by date, staff, action)
- 6 - Analytics (length of stay, common diagnoses)
- 7 - Patient admission form (diagnosis, vitals)
- 8 - Patient list (my patients, current bed, status)
- 9 - Vitals update form

Queue Algorithm:

Function: sort_waiting_queue(patients)

FOR each patient in waiting_queue:

severity_score = RED(3) > YELLOW(2) > GREEN(1)

wait_time_minutes = current_time - admission_time

deterioration_rate = (O2_sat_now - O2_sat_1h_ago) / 60 minutes

priority_score = (severity_score * 100) + wait_time_minutes + (deterioration_rate * 50)

SORT by priority_score DESC

RETURN ranked_queue

Severity Classifier: Rule based

Function: classify_severity(O2_sat, respiratory_rate, temp, has_comorbidities, age)

```
IF O2_sat < 88:
    RETURN "ICU"
ELIF O2_sat 88-94 AND (respiratory_rate > 25 OR age > 65 OR has_comorbidities):
    RETURN "ICU"
ELIF O2_sat 88-94:
    RETURN "Ward"
ELIF O2_sat > 95 AND respiratory_rate < 20:
    RETURN "Monitor_24h"
ELSE:
    RETURN "Ward"
```

Database:*Tables:*

hospitals (hospital_id, name, admin_email, total_icu_beds, total_ward_beds) - staff (staff_id, hospital_id, name, email, password_hash)

admins (admin_id, hospital_id, name, email, password_hash)

patients (patient_id, hospital_id, admission_date, diagnosis, vitals, severity_classification)

bed_assignments (assignment_id, patient_id, bed_number, bed_type, assigned_date, assigned_by_staff_id)

discharges (discharge_id, patient_id, discharge_date, reason, discharged_by_admin_id)

audit_log (log_id, timestamp, user_id, action, patient_id, details)

queue (queue_id, patient_id, hospital_id, priority_score, wait_time, deterioration_rate)